

Da Vinci Xi Tool Adapter and Electrosurgical Unit Interceptor

August 3, 2026

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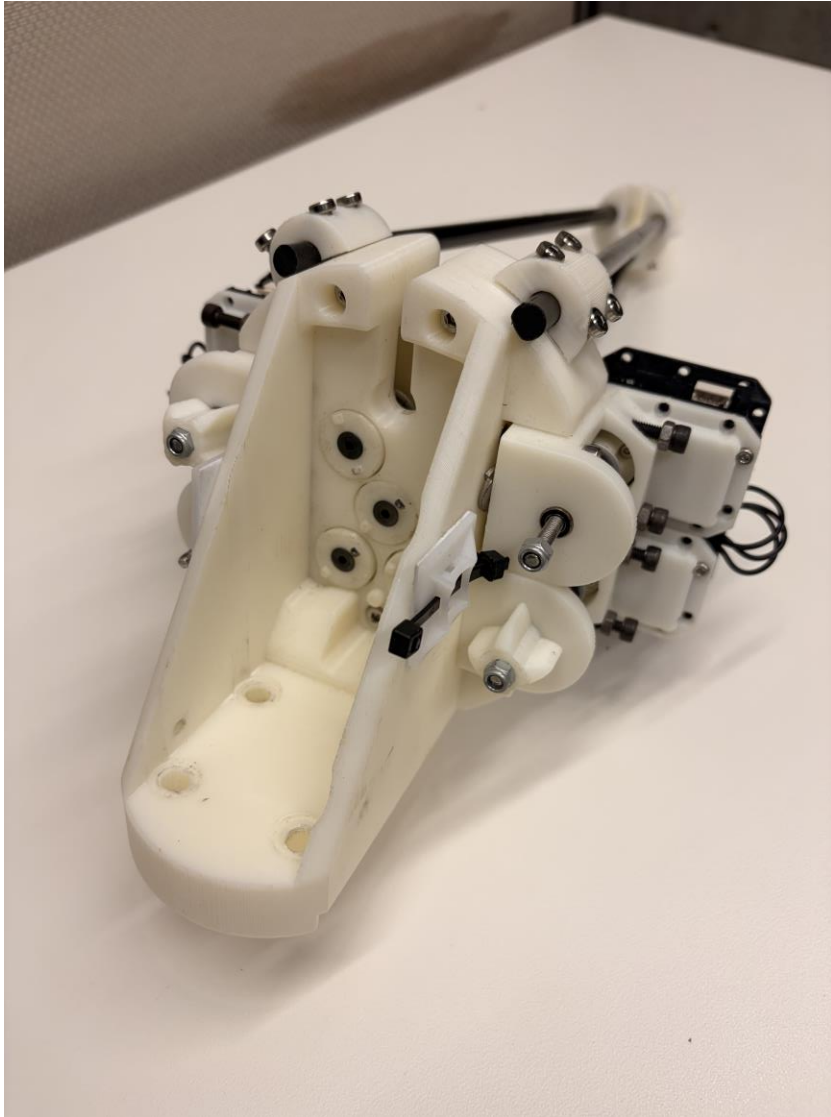


Da Vinci Xi Tool Adapter

Redesign a tool adapter for ARPA-H Partial Nephrectomy project



Old Adapter (MOPS)



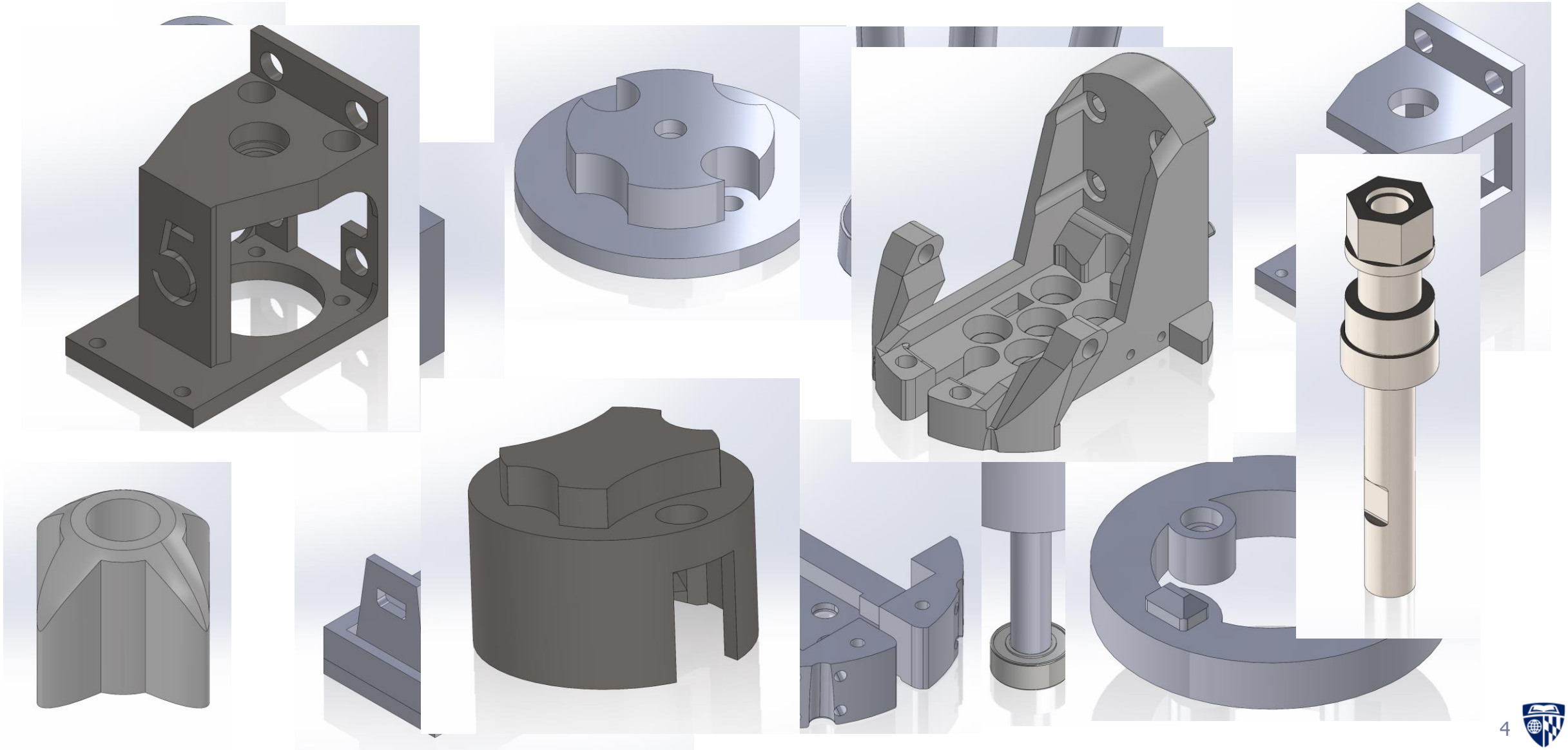
Based on MOPS (Modular and Open Platform for Surgical Robotics Research)

Major Version 2 motivations:

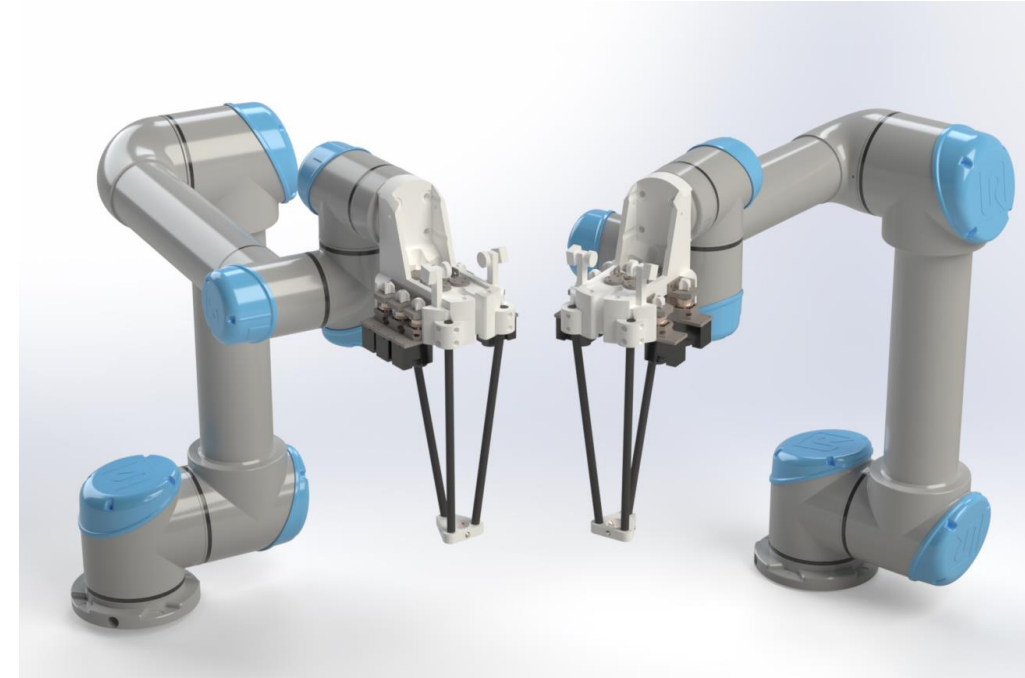
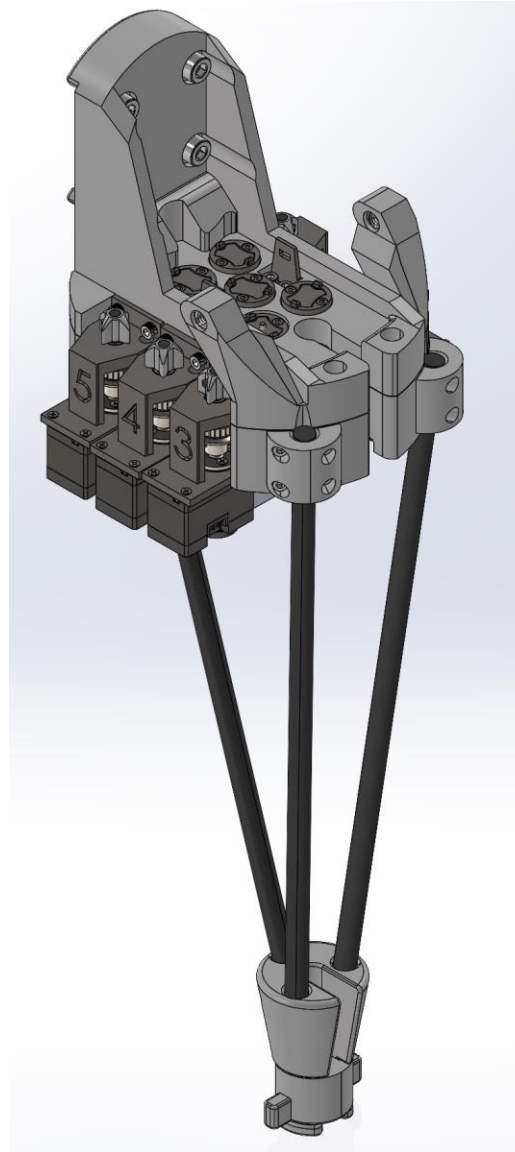
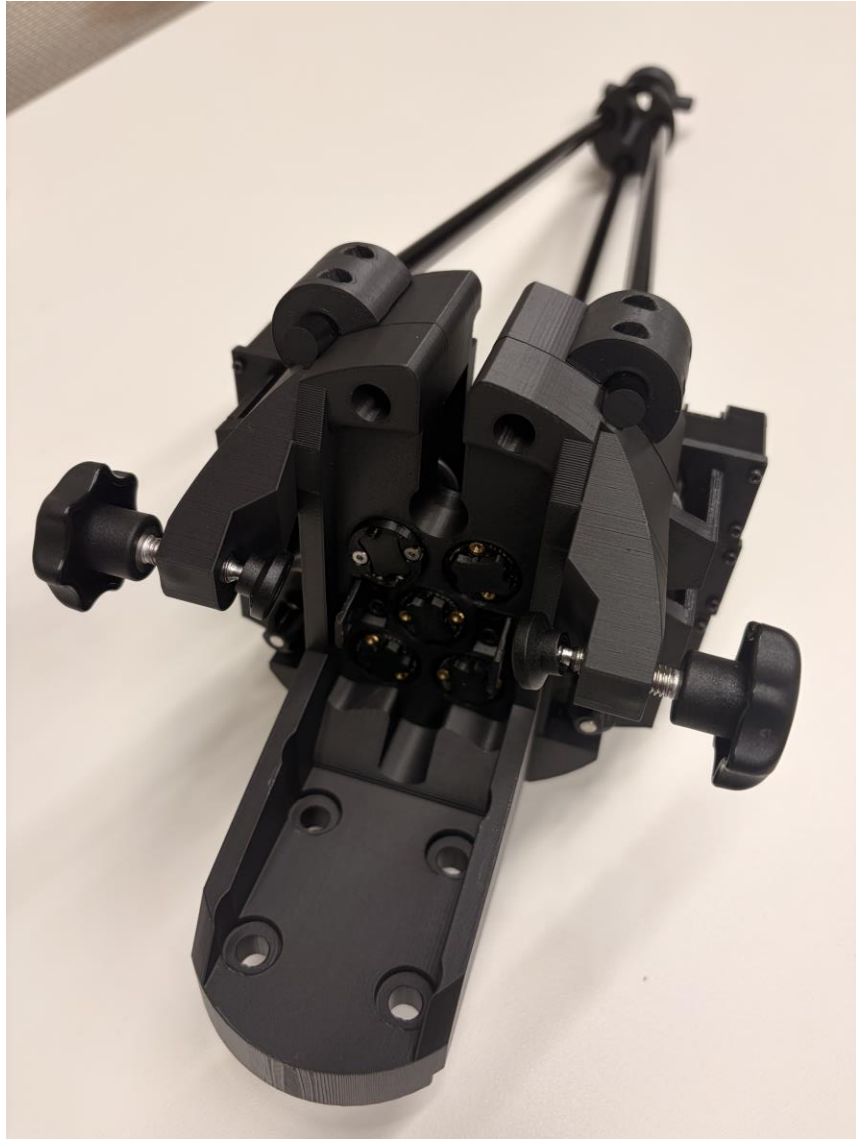
- Upgrade actuation
- Increase ease of connecting Xi tool
- Securely holding tool

K. L. Schwaner, I. Iturrate, J. K. Holm Andersen, C. Rosendahl Dam, P. T. Jensen and T. Rajeeth Savarimuthu, "MOPS: A Modular and Open Platform for Surgical Robotics Research," 2021 International Symposium on Medical Robotics (ISMR), Atlanta, GA, USA, 2021, pp. 1-8, doi: 10.1109/ISMR48346.2021.9661539.

Prototype Iterations



New Adapter



Motivation 1 - Actuation

- Goal: Incorporate fifth motor and reduce size

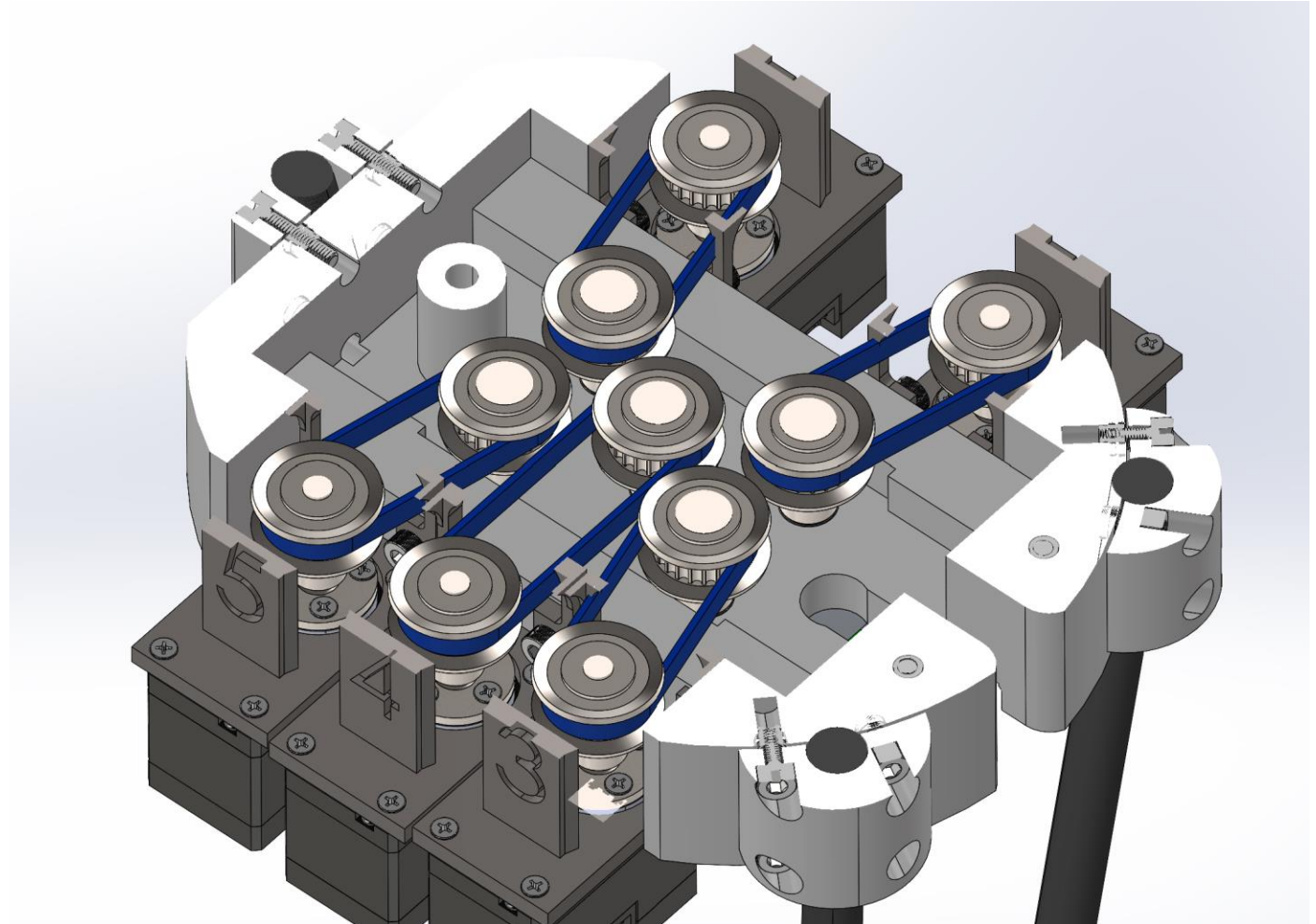
	Original (Dynamixel MX-28T)	New (Dynamixel XC330-M288-T)	
Dimensions	(35.6 x 50.6 x 35.5) = 63944mm	(20 x 34 x 26) = 17680mm ³	72% smaller
Weight	77g	23g	58% lighter
Control Modes	4	6	Current control
Stall Torque	1.5 Nm	0.93 Nm	
No Load RPM	59 RPM	65 RPM	



Motivation 1 - Actuation

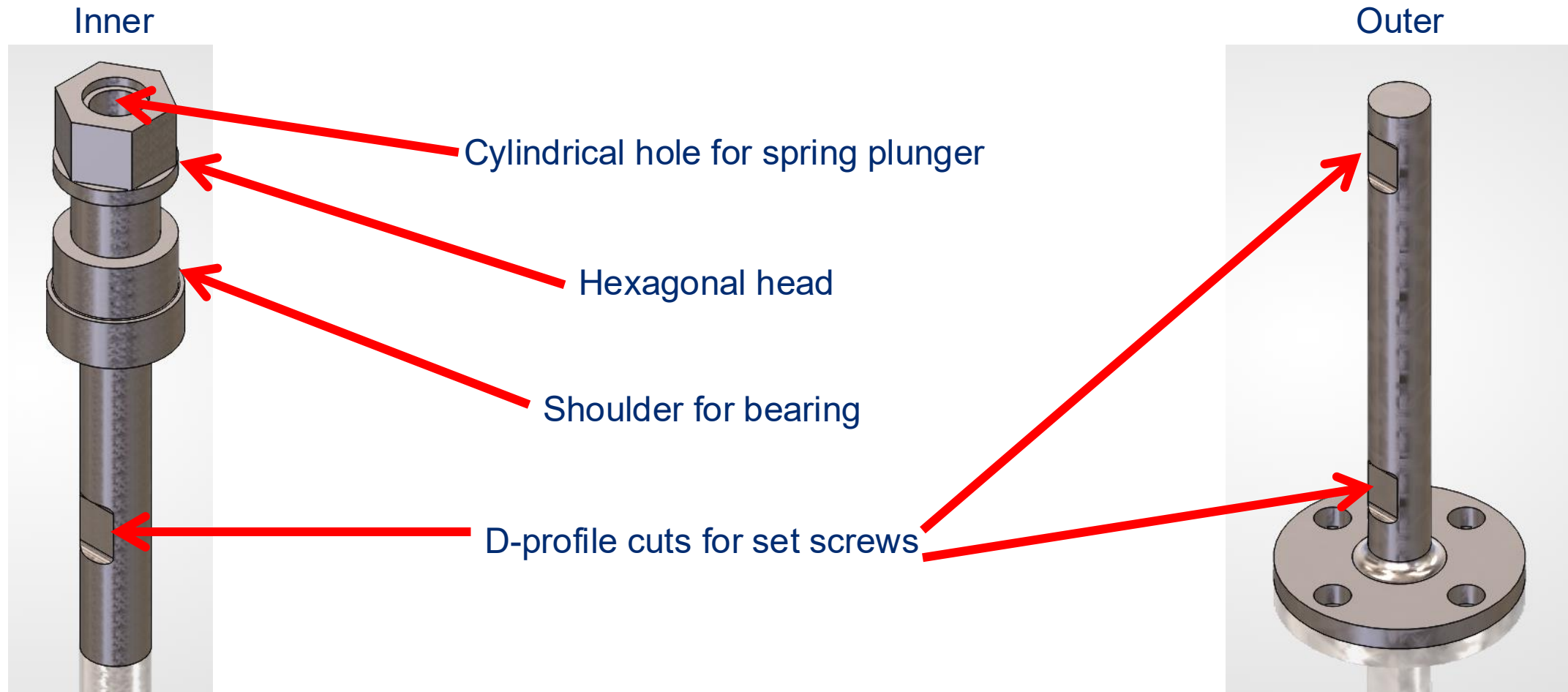
Goal: Shrink timing belt pulley system

- 22 teeth pulleys
- 106mm and 128mm timing belts
 - Tension considerations
- Ball bearings



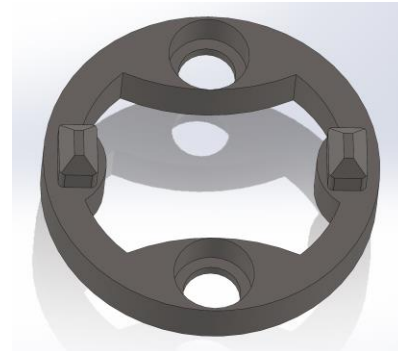
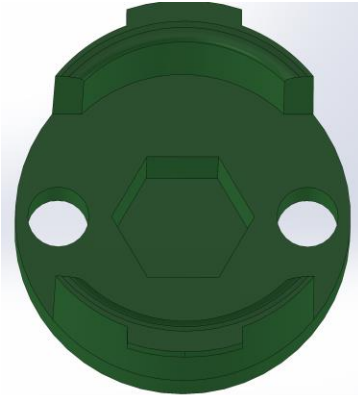
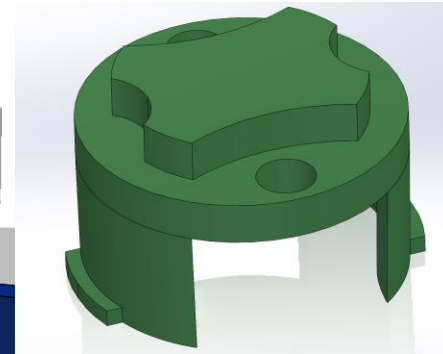
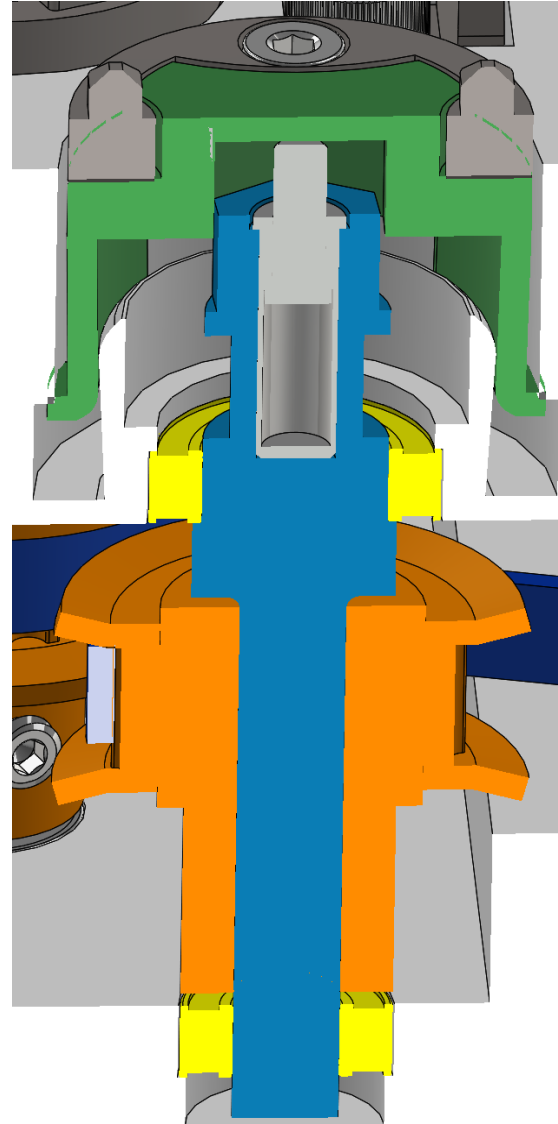
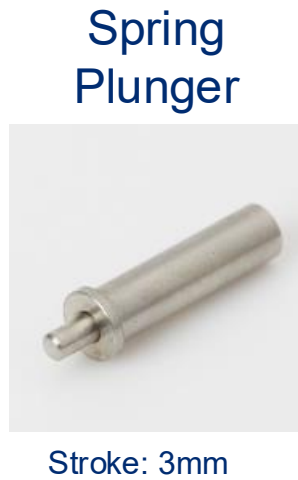
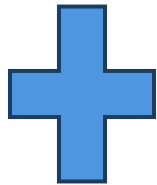
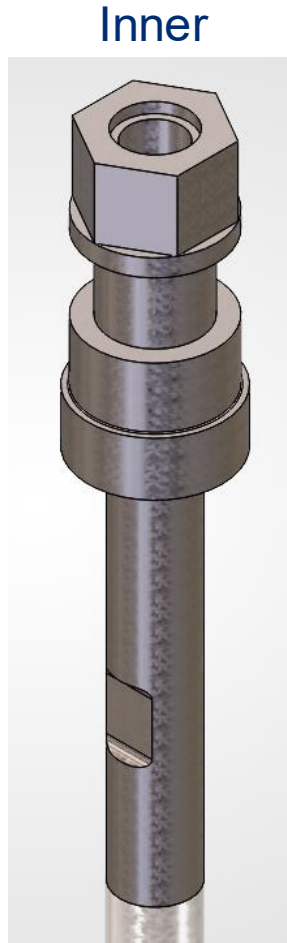
Motivation 2 – Ease of Tool Engagement

Goal: Incorporate a spring component (slide 1/2)



Motivation 2 – Ease of Tool Engagement

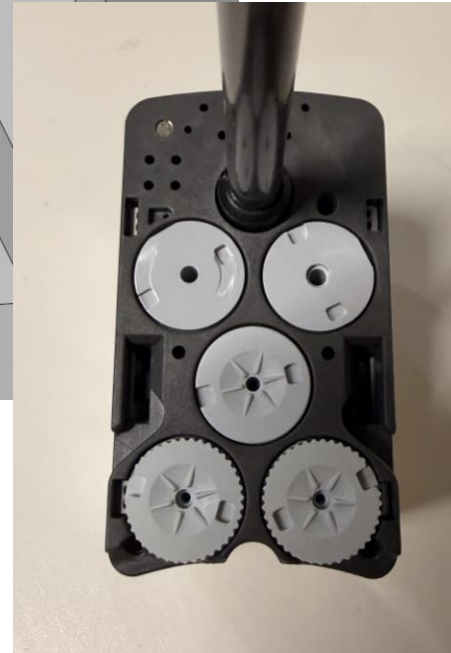
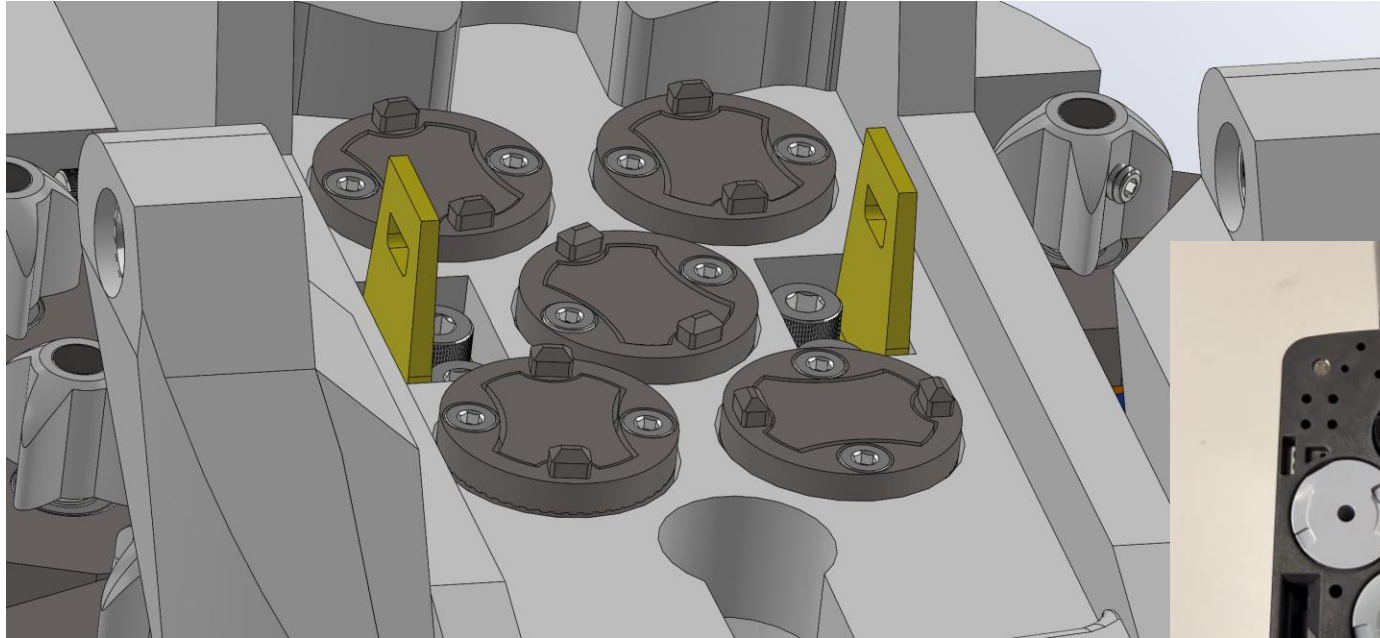
Goal: Incorporate a spring component (slide 2/2)



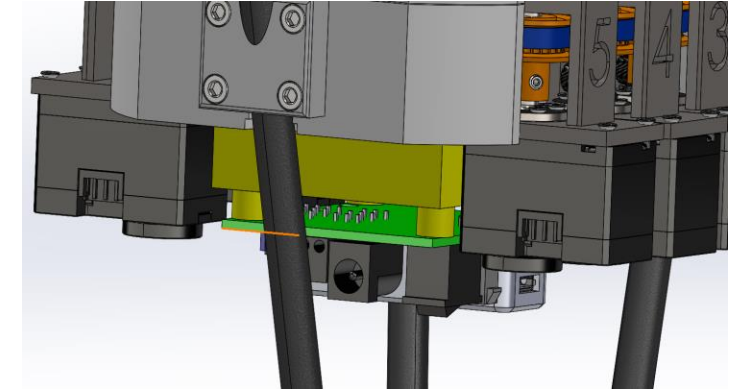
Motivation 3 – Secure Attachment

- Goal: Maintain precise positioning of tool tip

Snap clips



Control Board



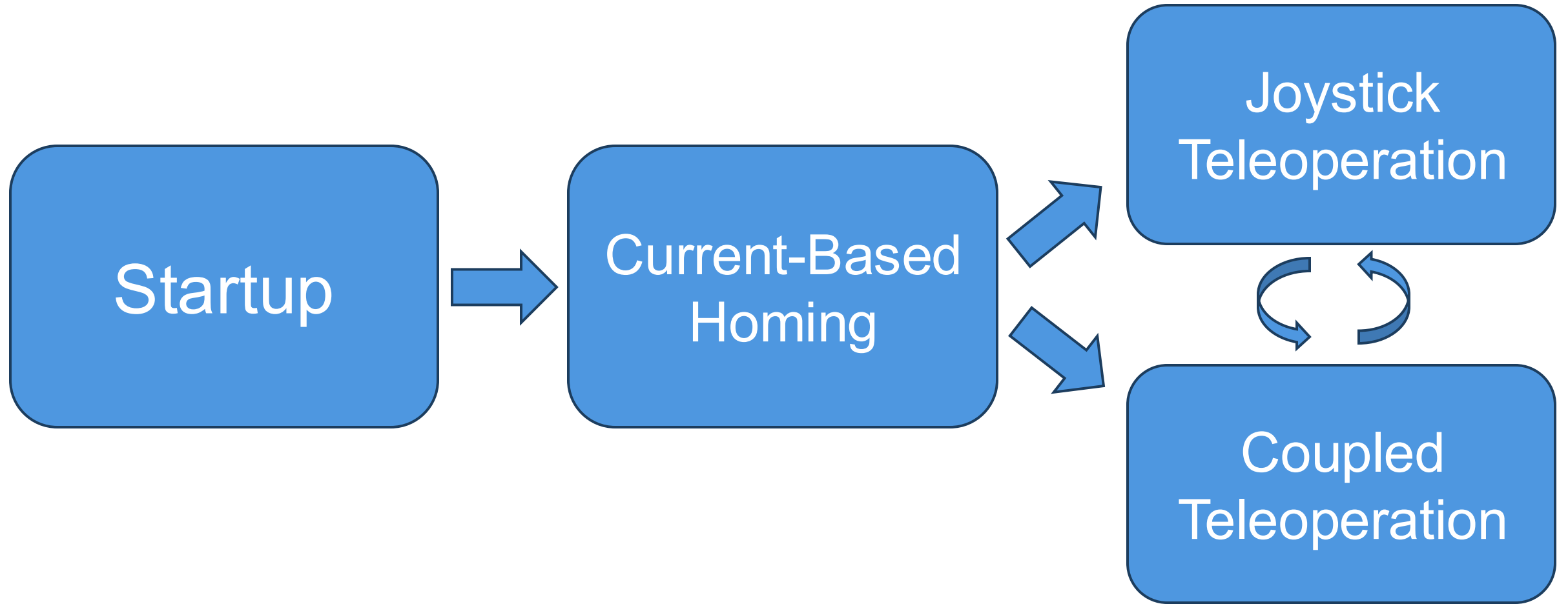
Star Knobs



Final Design



Software Modes



Startup

- 1) Confirm motor functionality
- 2) Enable torque
- 3) Move to optimal positions
- 4) Wait for tool



MCS



Forceps



```
tion-PC:~/dynamixels$ ros2 run dxl_joy_control dxl_joy_node
[INFO] [1785764525.469792735] [dxl_joy_node]: Position control activate, torque
enabled
[INFO] [1785764525.549746518] [dxl_joy_node]: Moving to starting locations for 5
seconds
[INFO] [1785764530.574783057] [dxl_joy_node]: Put the Xi tool on the adapter, an
d press the 'A' button on the joy controller
```

Current-Based Homing

- Rotate both directions
- Record limits
 - Threshold current peaks
- Rotate to the midpoint

```
[INFO] [1784553805.368524444] [dxl_joy_node]: Motor 1 position: 50379 and current: 125 mA
[INFO] [1784553805.624362201] [dxl_joy_node]: Motor 1 position: 50581 and current: 122 mA
[INFO] [1784553805.864378682] [dxl_joy_node]: Motor 1 position: 50776 and current: 126 mA
[INFO] [1784553806.120469199] [dxl_joy_node]: Motor 1 position: 50974 and current: 122 mA
[INFO] [1784553806.376334539] [dxl_joy_node]: Motor 1 position: 51169 and current: 121 mA
[INFO] [1784553806.616521757] [dxl_joy_node]: Motor 1 position: 51332 and current: 220 mA
[INFO] [1784553806.872373501] [dxl_joy_node]: Motor 1 position: 51432 and current: 499 mA
[INFO] [1784553807.128330418] [dxl_joy_node]: Motor 1 position: 51755 and current: 102 mA
[INFO] [1784553807.368312916] [dxl_joy_node]: Motor 1 position: 51966 and current: 89 mA
[INFO] [1784553807.624405912] [dxl_joy_node]: Motor 1 position: 52102 and current: 22 mA
```



Teleoperation

Coupling Matrix

```
t0 = 1.5  
t1 = 1.5632  
t2 = 1.0186  
t3 = 1.2177  
w = 0.6089
```

```
self.K = t0 * np.array([  
    [ t1,  0,    0,    0 ],  
    [ 0,  t2,    0,    0 ],  
    [ 0,  t2*w, -t3,  0 ],  
    [ 0, -t2*w,  0,  t3 ],  
])
```

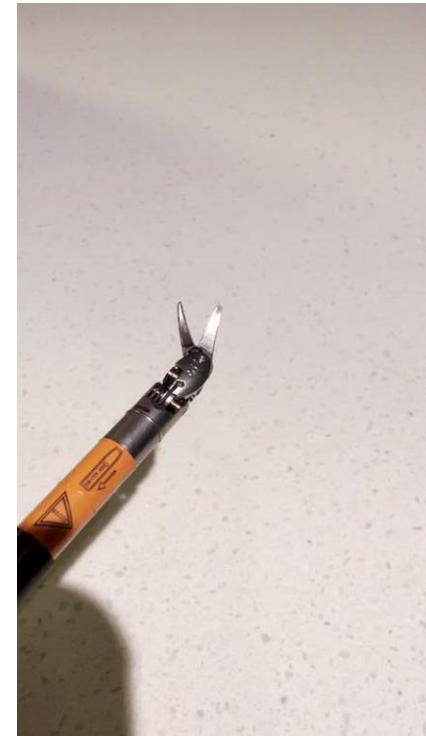
Roll



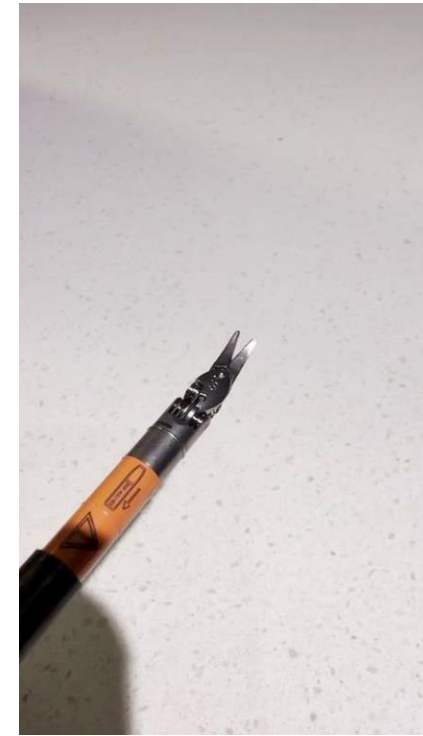
Pitch



Yaw



Scissors



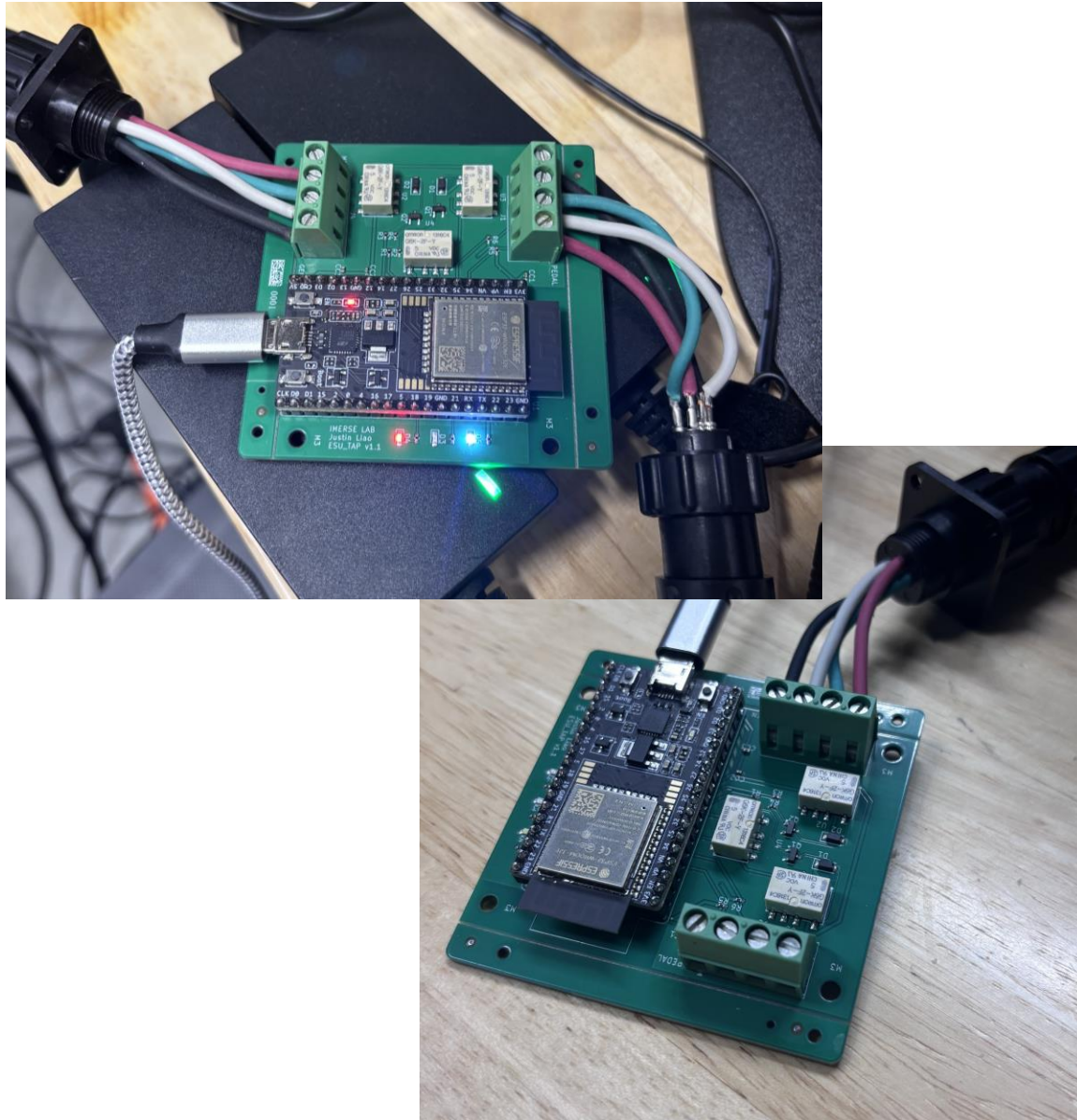
Next Steps: Continued ROS2 integration and usage on Vention platform

Electrosurgical Unit Interceptor

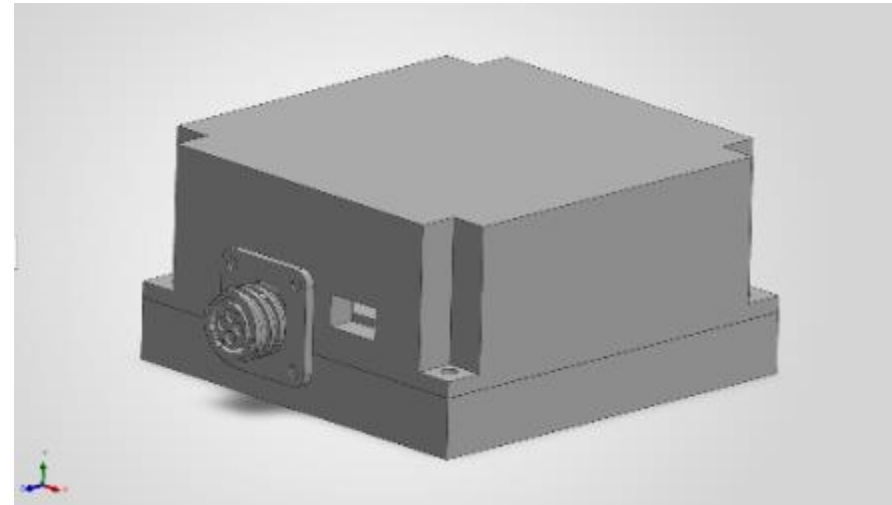
Autonomously enable and log activated generator functions



Monopolar Control Board



- 1) Automatic fail-safe
- 2) LED indicators
- 3) Microros on ESP32
- 4) Publishes state on ros topic
- 5) Intuitive design



Results

```

justin@blue: ~-100x27
leaving...
Hard resetting via RTS pin...
===== [SUCCESS] Took 6.72 seconds =====
justin@blue:~/ESU_TAP/microros_firmware$ docker run -it --rm -v /dev/dev --privileged --net=host
t \
  microros/micro-ros-agent:humble serial --dev /dev/ttyUSB0 -b 115200
[178535397.716206] info | Root.cpp | set_verbose_level | logger setup
| verbose_level: 4
[178535398.291123] info | Root.cpp | create_client | create
| client_key: 0x1D442DC4, session_id: 0x81
[178535398.291166] info | SessionManager.hpp | establish_session | session established
| client_key: 0x1D442DC4, address: 0
[178535398.308602] info | ProxyClient.cpp | create_participant | participant created
| client_key: 0x1D442DC4, participant_id: 0x000(1)
[178535398.326207] info | ProxyClient.cpp | create_topic | topic created
| client_key: 0x1D442DC4, topic_id: 0x000(2), participant_id: 0x000(1)
[178535398.337093] info | ProxyClient.cpp | create_publisher | publisher created
| client_key: 0x1D442DC4, publisher_id: 0x000(3), participant_id: 0x000(1)
[178535398.349153] info | ProxyClient.cpp | create_datawriter | datawriter created
| client_key: 0x1D442DC4, datawriter_id: 0x000(5), publisher_id: 0x000(3)
[178535398.374346] info | ProxyClient.cpp | create_replier | replier created
| client_key: 0x1D442DC4, requester_id: 0x000(7), participant_id: 0x000(1)
[178535398.400431] info | ProxyClient.cpp | create_replier | replier created
| client_key: 0x1D442DC4, requester_id: 0x001(7), participant_id: 0x000(1)

justin@blue: ~-100x27
g ENTER -> toggle COAG (g = coaG, since c is taken)
x ENTER -> all OFF (safety)
s ENTER -> show current state
h ENTER -> help
Ctrl+C -> quit (deactivates everything)

Tip: bare ENTER toggles the default channel (cut), matching the old behaviour.
> c
[INFO] [1785355730.719043997] [pedal_commander]: CUT ON -> ok (cut on)
> c
[INFO] [1785355736.566605950] [pedal_commander]: CUT OFF -> ok (cut off)
> g
[INFO] [1785355736.975184005] [pedal_commander]: COAG ON -> ok (coag on)
> x
[INFO] [1785355737.961999633] [pedal_commander]: CUT OFF -> ok (cut off)
[INFO] [1785355737.978910800] [pedal_commander]: COAG OFF -> ok (coag off)
> c
[INFO] [1785355743.942663221] [pedal_commander]: CUT ON -> ok (cut on)
> g
[INFO] [1785355745.038705230] [pedal_commander]: COAG ON -> ok (coag on)
> s
state: CUT=ON COAG=ON
> x
[INFO] [1785355751.422003272] [pedal_commander]: CUT OFF -> ok (cut off)
[INFO] [1785355753.422609270] [pedal_commander]: COAG OFF -> ok (coag off)
>

justin@blue: ~-ESU_TAP/101x27
[INFO] [1785355809.030193489] [pedal_logger]: [computer] cut release @ 1785355809029957436 ns (cut=0, coag=0)
[INFO] [1785355809.230017674] [pedal_logger]: [computer] cut press @ 1785355809229708233 ns (cut=1, coag=0)
[INFO] [1785355809.255295661] [pedal_logger]: [computer] cut release @ 1785355809255060370 ns (cut=0, coag=0)
[INFO] [1785355809.475480275] [pedal_logger]: [computer] cut press @ 1785355809475200219 ns (cut=1, coag=0)
[INFO] [1785355809.503210404] [pedal_logger]: [computer] cut release @ 1785355809503022712 ns (cut=0, coag=0)
[INFO] [1785355809.697524628] [pedal_logger]: [computer] cut press @ 1785355809697207253 ns (cut=1, coag=0)
[INFO] [1785355809.731882886] [pedal_logger]: [computer] cut release @ 1785355809731631825 ns (cut=0, coag=0)
[INFO] [1785355809.941480702] [pedal_logger]: [computer] cut press @ 1785355809941165140 ns (cut=1, coag=0)
[INFO] [1785355809.976216408] [pedal_logger]: [computer] cut release @ 1785355809975985394 ns (cut=0, coag=0)
[INFO] [1785355810.201799072] [pedal_logger]: [computer] cut press @ 1785355810201517153 ns (cut=1, coag=0)
[INFO] [1785355810.259047456] [pedal_logger]: [computer] cut release @ 1785355810258005162 ns (cut=0, coag=0)
[INFO] [1785355810.456613233] [pedal_logger]: [computer] cut press @ 1785355810456337716 ns (cut=1, coag=0)
[INFO] [1785355810.503529127] [pedal_logger]: [computer] cut release @ 1785355810503292904 ns (cut=0, coag=0)

justin@blue: ~-101x27
ROUND-TRIP (call -> response) (n=50)
=====
min 16.04 ms
mean 17.39 ms
median 16.63 ms
p95 25.50 ms
p99 25.96 ms
max 25.97 ms
stdev 2.55 ms

COMMAND (call -> observed state) (n=50)
=====
min 24.97 ms
mean 38.79 ms
median 35.01 ms
p95 68.72 ms
p99 83.30 ms
max 90.30 ms
stdev 14.91 ms

Mean COMMAND - mean ROUND-TRIP = 21.39 ms
Expected ~5.0 ms from the 10 ms publish timer alone. Much larger means executor scheduling overhead;

justin@blue: ~-101x27
ROUND-TRIP (call -> response) (n=99)
=====
min 14.96 ms
mean 17.38 ms
median 16.61 ms
p95 25.47 ms
p99 26.11 ms
max 29.34 ms
stdev 2.67 ms

COMMAND (call -> observed state) (n=99)
=====
min 25.25 ms
mean 41.44 ms
median 36.13 ms
p95 83.41 ms
p99 109.03 ms
max 134.72 ms
stdev 20.05 ms

Mean COMMAND - mean ROUND-TRIP = 24.06 ms
Expected ~5.0 ms from the 10 ms publish timer alone. Much larger means

```

- Round-trip: **16.6 ms** median
- Command latency: **36 ms** median (p99: 109 ms)
- Sample size: n=100
- 6–8× faster than human reaction time (~250 ms)
- Can determine surgeon/computer

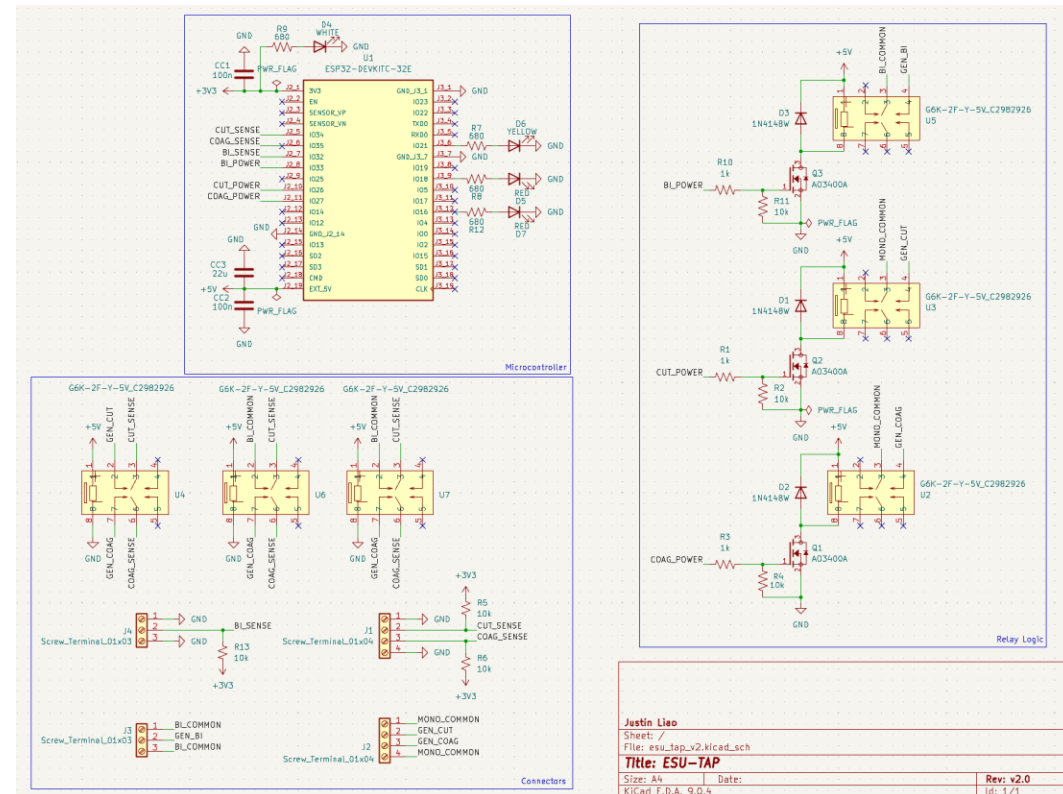


Future Work

Additional safety features

- Bypass bank
- Watchdog-held coil drive
- SD Drive logging
- FCFS mutual exclusion

- V2 (Monopolar + Bipolar)
- V3 (Monopolar + Bipolar + Ligasure)



Thank you!
Questions?

